



Technical Reproducibility Validation Report

STRait Razor

Version v3

STR / SNP panel

Verification date	2026-08-01
Repository commit	b618e9345ab40f348b504083ae8de2b39abb60fa
Attestation level	Runs + Output Structure Validation
Permalink	https://strhub.app/verified/strait-razor-PowerSeqv2.31
STRhub	https://strhub.app

SCOPE OF THIS REPORT

Independent automated verification of reproducible execution. It confirms the tool installs, runs end-to-end, and produces structurally valid output in a standardized environment.

This is **not** an analytical validation. Genotype accuracy, concordance, forensic suitability, and regulatory compliance are out of scope.

WHAT WAS VERIFIED

- Source available
- Installation successful
- End-to-end execution
- Output generated

NOT VERIFIED

- Genotype accuracy
- Concordance
- Forensic validity
- Regulatory compliance



1. Executive Summary

PURPOSE	Verify that the tool installs, runs end-to-end, and produces structurally valid output.
RESULT	5/5 gates passed. Runs + Output Structure Validation
ATTESTATION LEVEL	Runs + Output Structure Validation
SCOPE	Installation, execution, and output structure verification.
NOT EVALUATED	Genotype accuracy · Concordance · Forensic validity · Regulatory compliance

2. Reproducibility Metadata

Tool	STRait Razor
Version	v3
Repository	https://github.com/Ahhgust/STRaitRazor
Commit (pinned)	b618e9345ab40f348b504083ae8de2b39abb60fa
Container OS	ubuntu-22.04
Marker panel	STR / SNP panel
Verified on	2026-08-01 08:20:43 UTC
CI run	https://github.com/Tfronta/strhub-verified/actions/runs/30691557482

3. Exact Run Command

The following command was executed verbatim in the CI environment. Replacing the input paths under /data/in with your own inputs reproduces the verified run.

```
# Environment: ubuntu-22.04 · STRait Razor v3 · commit b618e93
$ str8rZR \
-c \
/opt/strait-razor/PowerSeqv2.31.config \
/data/in/sample.fastq \
> \
/data/out/sample.allsequences.txt
```

4. Verification Gates

Available	The pinned public source exists	PASS
Installs	The environment builds from source	PASS
Runs	Executes end-to-end without crashing	PASS
Expected IO	Produces a non-empty file in the declared format	PASS

Output Structure Validation	Output structure matches expected genotype-bearing format	PASS
Execution log	—	

5. Out of Scope

This report does not evaluate any of the following:

- Genotype correctness or accuracy
- Concordance against known truth sets
- Sensitivity, specificity, or stutter performance
- Allele calling accuracy or forensic casework suitability
- Regulatory compliance or ISO accreditation
- Multi-laboratory or multi-dataset reproducibility

6. Output Content Evidence

Output file	sample.allsequences.txt
Format	TSV
Sequence records	1172
Distinct STR loci	44
Total reads	4,634
Max depth (single locus)	179

7. Verification Data

This tool does not include its own demo or test data. STRhub ran the verification using the public reference dataset listed below.

Dataset	NIST mds2-2157, Illumina STR (ForenSeq slice, donor NTD01)
Source	https://data.nist.gov/od/id/mds2-2157
License	research / training / education only (per NIST); not for donor identification or database searching
Ref. genome	—
Loci tested	Amelogenin · CSF1PO · D10S1248 · D12S391 · D13S317 · D16S539 · D18S51 · D19S433 · D1S1656 · D21S11 · D22S1045 · D2S1338 · D2S441 · D3S1358 · D5S818 · D7S820 · D8S1179 · DYS19 · DYS385 · DYS389II · DYS390 · DYS391 · DYS392 · DYS393 · DYS437 · DYS438 · DYS439 · DYS448 · DYS456 · DYS458 · DYS481 · DYS533 · DYS549 · DYS570 · DYS576 · DYS635 · DYS643 · FGA · PentaD · PentaE · TH01 · TPOX · Y-GATA-H4 · vWA

8. Verification Matrix

PASS	Reference dataset	NIST mds2-2157, Illumina STR (ForenSeq slice, donor NTD01)
	README check (5/5)	Advisory: all items present
	Install / environment setup	Present
	Run command	Present
	Expected input format	Present
	Produced output format	Present
	Dependencies / versions	Present

9. Limitations

- Single reference dataset per input type
- Single containerized environment (Docker / ubuntu-22.04)
- No truth-set comparison or ground-truth genotypes
- No accuracy or concordance assessment
- No forensic validation of results
- Short-read limitations apply (very long STR alleles may not span reads)

10. Scope and Disclaimers

Executed end-to-end in the stated environment with output in the expected format. Concerns reproducible execution only; no claim of accuracy, casework fitness, or regulatory validation.

Each result is a dated snapshot, verified on the tool's public repository at a pinned commit. STRhub does not store tool source code.

11. Conclusion

◆ Runs end-to-end

STRait Razor v3 installs and executes without error in a clean ubuntu-22.04 environment at the pinned commit. All 5 verification gates were passed.

◆ Produces valid output

The tool generated a structurally valid TSV output file with genotype calls across 44 target forensic STR loci, with a maximum read depth of 179 at DYS439.

◆ No bundled demo data

STRait Razor does not include its own test or demo dataset. STRhub Verified supplied NIST mds2-2157, Illumina STR (ForenSeq slice, donor NTD01) as the reference input for this verification run. Users replicating this result should use the same or equivalent publicly available reference material.

◆ Reproducibility statement

The exact command reported in Section 3, executed at the pinned commit in the described environment, is sufficient to reproduce this verification result independently.



Appendix: Detected Markers with Read Depth

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LOCUS	READ DEPTH		LOCUS	READ DEPTH	
DYS439	205	████████████████████	D5S818	96	████████
DYS385	204	████████████████████	DYS19	93	████████
D2S1338	200	████████████████████	TH01	90	████████
DYS393	177	██████████████████	DYS549	89	██████
Amelogenin	170	██████████████████	DYS458	88	██████
D10S1248	151	██████████████	Y-GATA-H4	88	██████
D19S433	136	██████████	DYS533	87	██████
D21S11	131	██████████	D16S539	86	██████
D3S1358	131	██████████	D13S317	84	██████
DYS481	128	██████████	DYS456	83	██████
D8S1179	127	██████████	PentaE	82	██████
DYS448	125	██████████	vWA	79	██████
D7S820	121	██████████	DYS576	74	████
FGA	119	██████████	PentaD	69	████
TPOX	119	██████████	D12S391	67	████
DYS390	119	██████████	CSF1PO	64	████
DYS389II	107	████████	D2S441	63	████
DYS643	106	████████	DYS570	63	████
DYS391	105	████████	DYS392	59	████
DYS437	103	████████	D22S1045	58	████
DYS635	100	██████	D18S51	48	███
D1S1656	98	██████	DYS438	42	███